

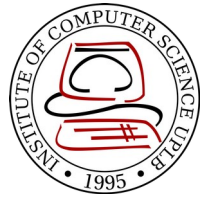
CMSC 137

Data Communications and Networking

Lecture Module
13 – Data Link Control

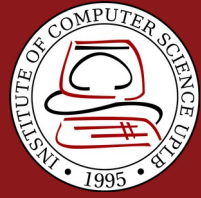


University of the Philippines
LOS BAÑOS



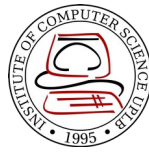
Module 13: Data Link Control

Rozano S. Maniaol
Lecturer



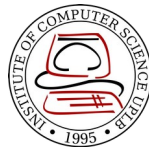
Framing

Framing



- Separates a message from one source to destination, or from other messages to other destinations by adding a **sender address** and a **destination address**
- Destination address tells where the frame is to go
- Sender address helps the recipient acknowledge the receipt
- A very large frame makes flow and error control very inefficient – a single bit error requires re-transmission

Framing

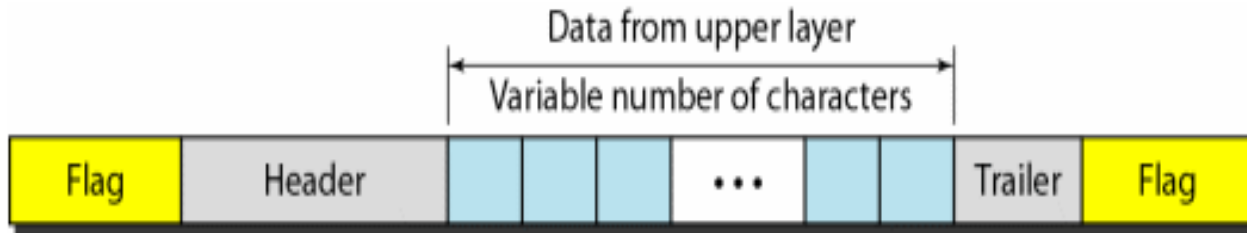


- **Fixed-size framing** – no need to define boundaries of the frames; ex. ATM networks
- **Variable-size framing** – need to define the end of the frame and beginning of the next; used in most networks
 - Character-oriented
 - Bit-oriented

Character-oriented Framing



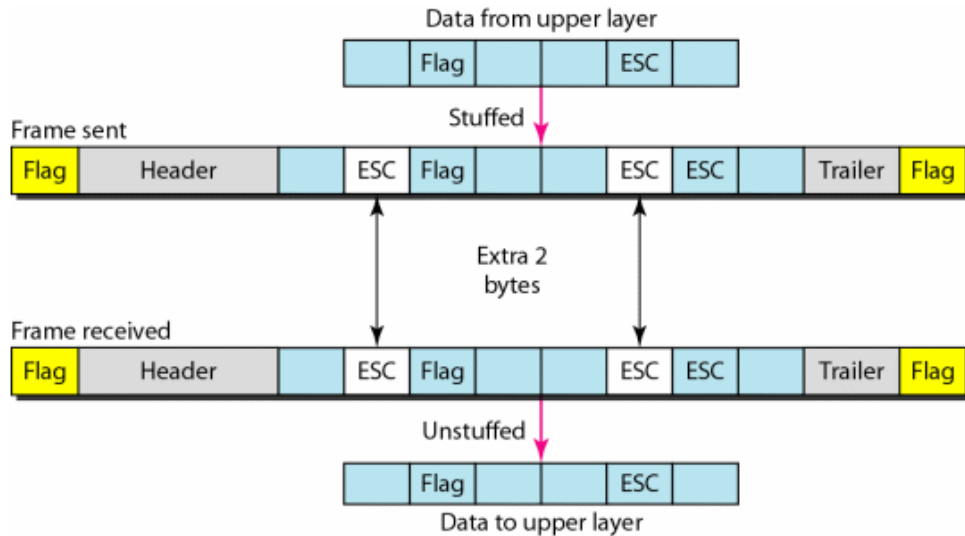
- Data to be carried are 8-bit characters from a coding system such as ASCII
- Header and trailer are multiples of 8 bits
- 8-bit (1 byte) flag is added at the beginning and end of a frame



Character-oriented Framing



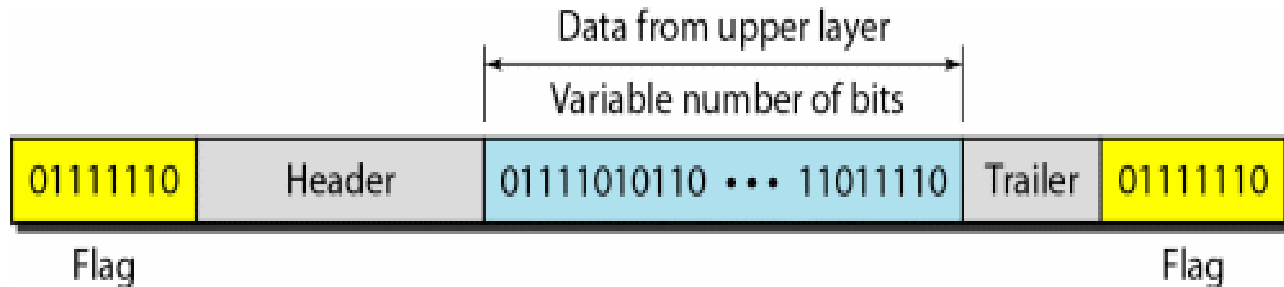
- **Byte-Stuffing** - Prevents data from being interpreted as flag
- A special byte, called escape character (ESC), is added to the data section of the frame when there is a character with the same pattern as the flag



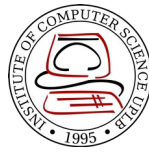
Bit-oriented Framing



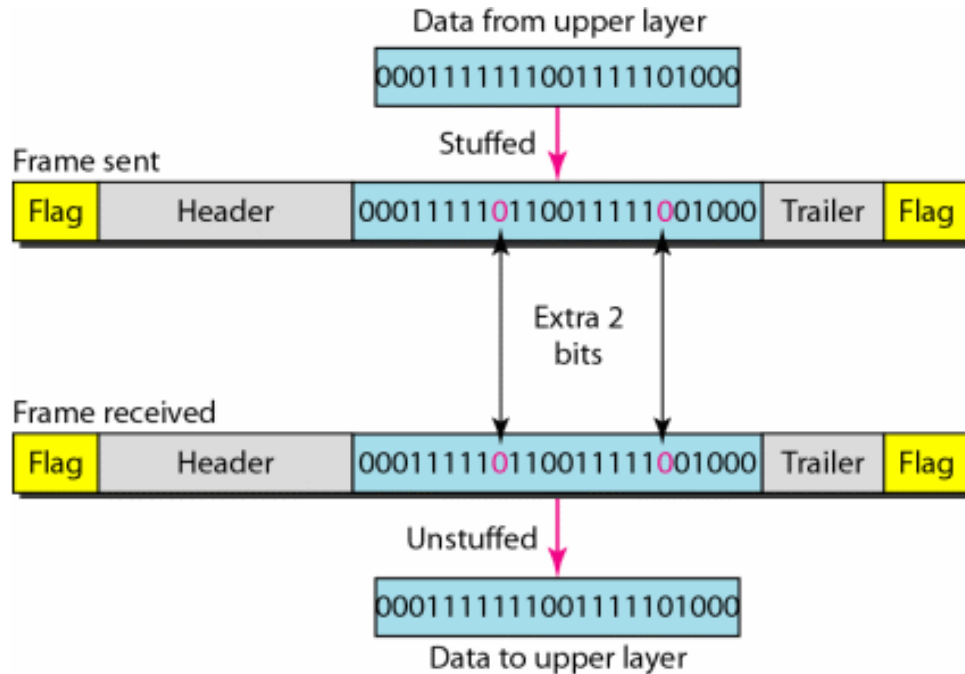
- Data section of frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, etc.
- Delimiter is a special bit pattern: 01111110

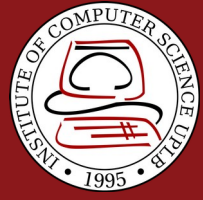


Bit-oriented Framing



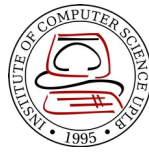
- Adds one extra 0 whenever five consecutive 1s follow a 0 in the data





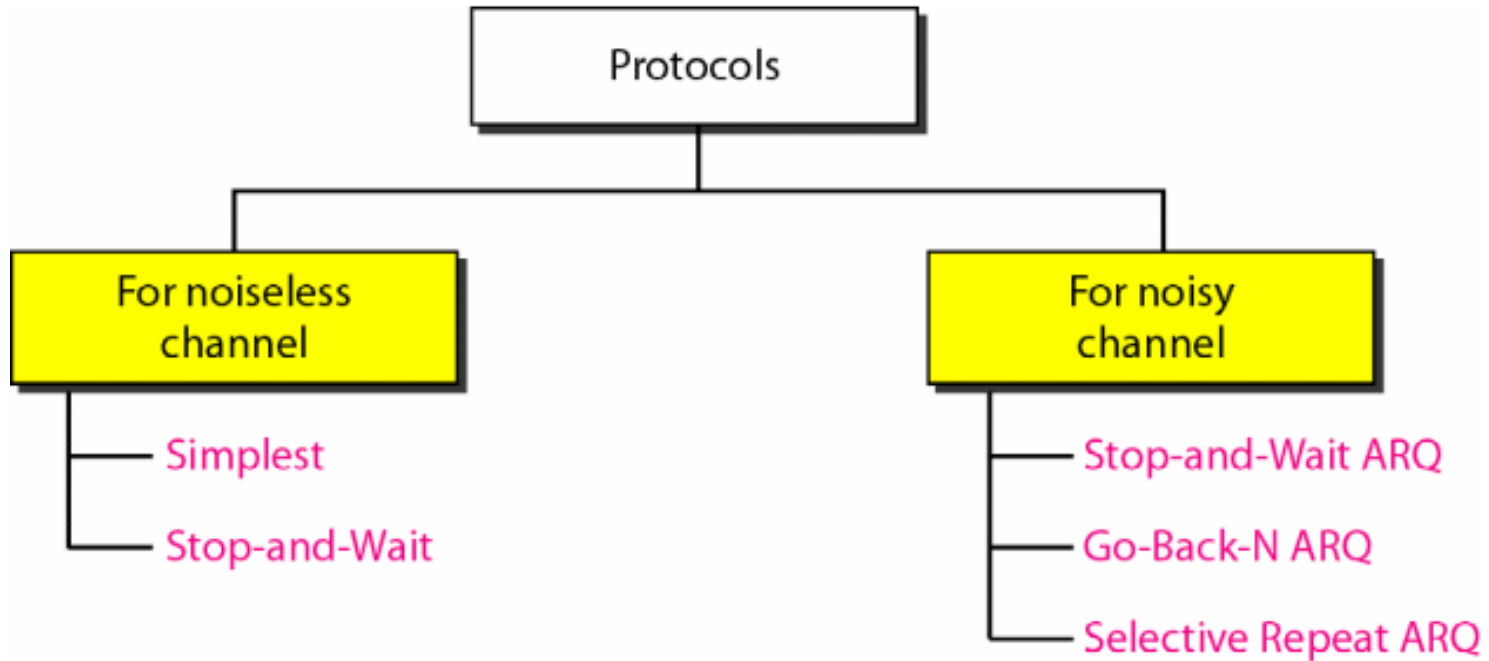
Flow and Error Control

Flow and Error Control

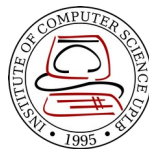


- Flow control coordinates the amount of data that can be sent before receiving an acknowledgment
- Error control is both error detection and correction
- **Automatic Repeat Request** (ARQ) – any time an error is detected in an exchange, specified frames are re-transmitted

Protocols

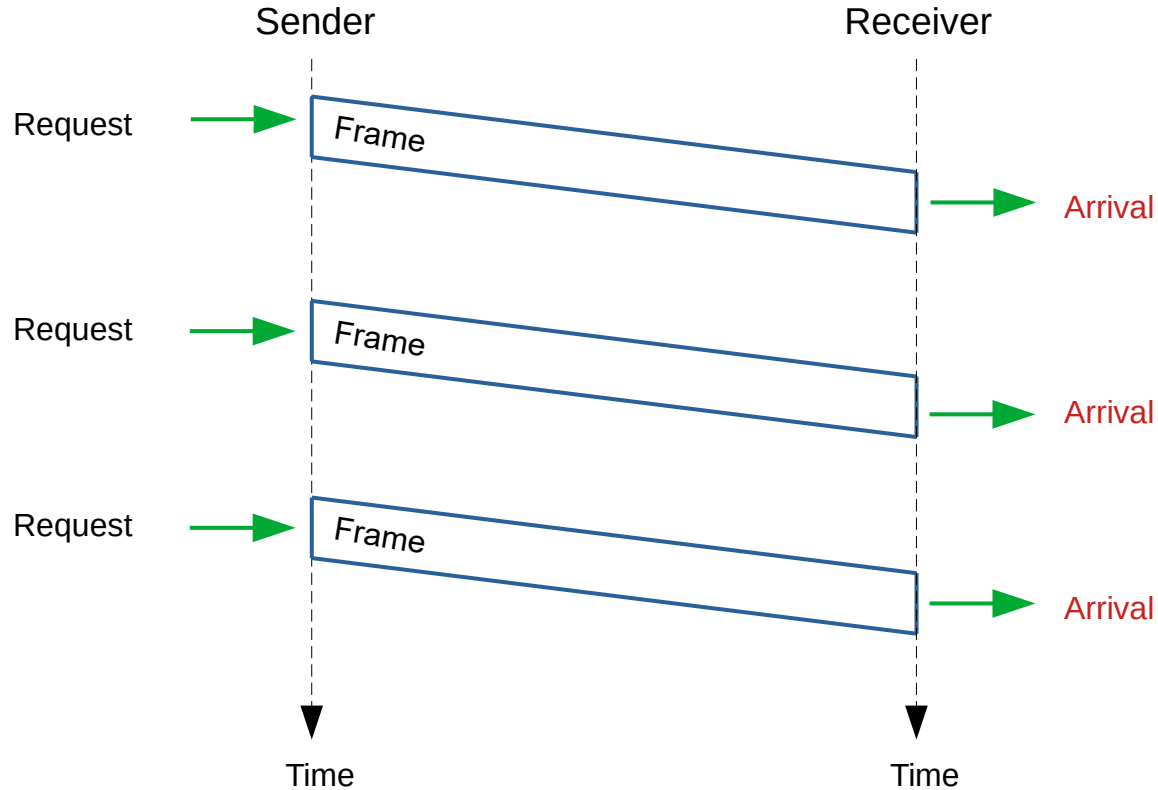
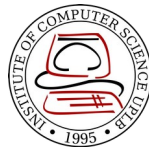


Protocols



- Note that the examples' focus are on unidirectional transfer
- Specialized frames
 - ACK – acknowledgment
 - NAK – negative acknowledgment
- **Piggybacking** – NAKs and ACKs included in the data frame
- Implemented in software

Simplest Protocol

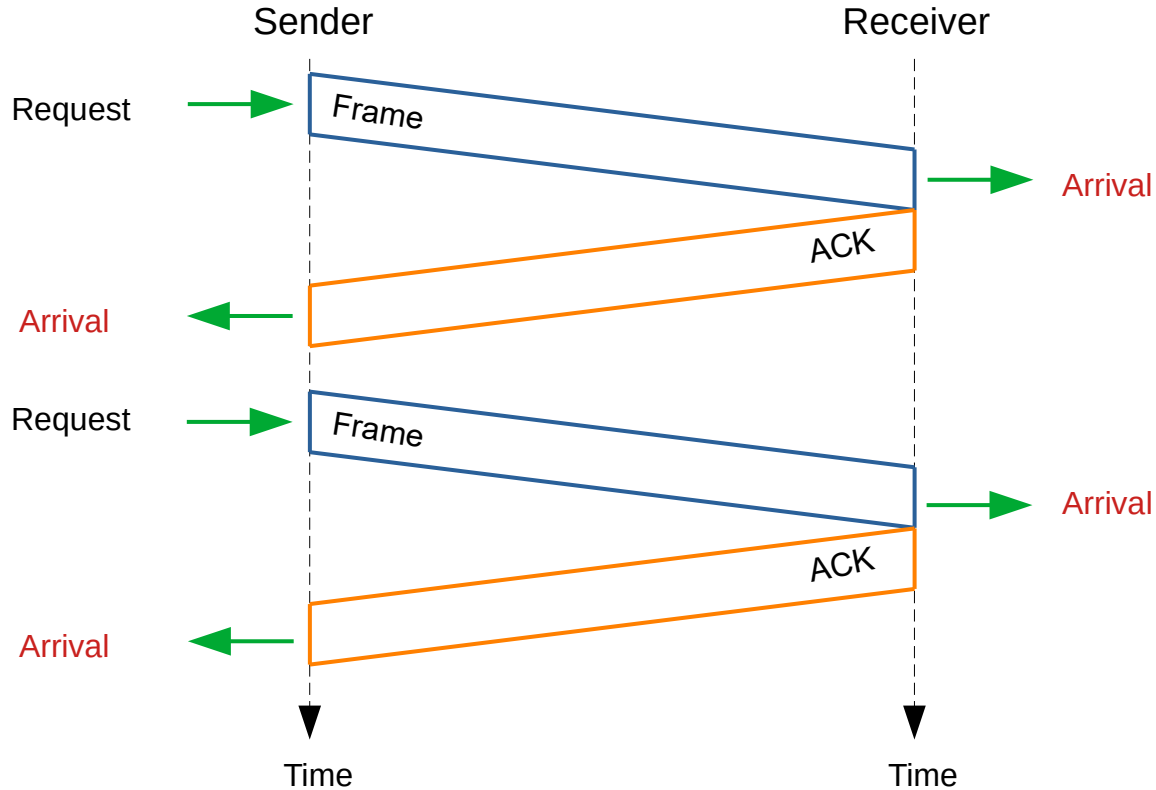
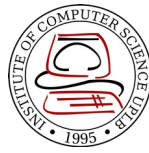


Stop and Wait Protocol

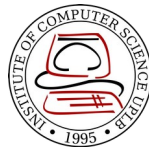


- If the data frames arrive at the receiver site faster than they can be processed, the frames must be stored until their use
- Sender sends one frame, until it receives confirmation from the receiver and then sends next frame

Stop and Wait Protocol

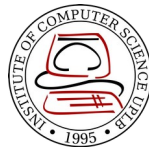


Stop and Wait ARQ



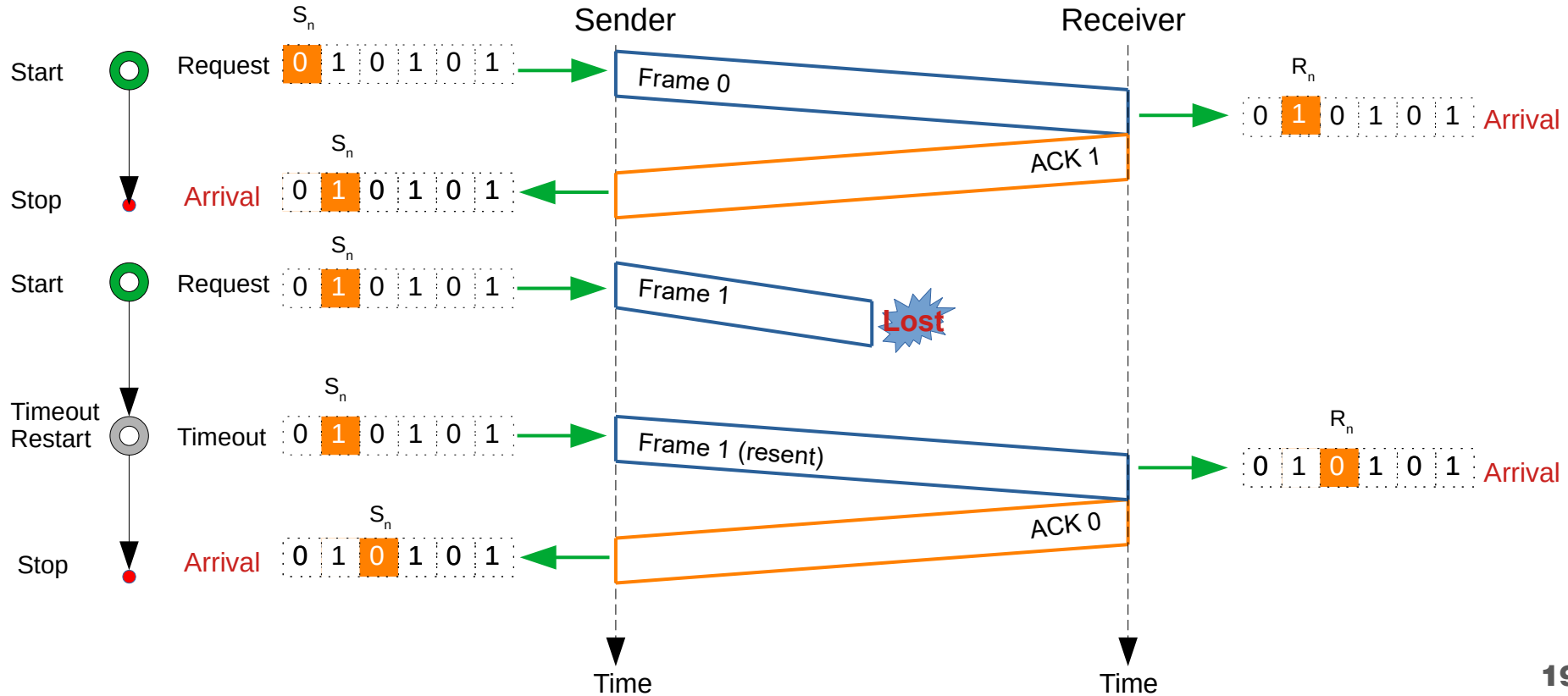
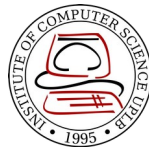
- Adds redundancy bits to data frame
- At the receiver end, if a frame is corrupted, it is **silently discarded**
- Numbering frames (**sequence numbers**) allows detection of lost frames
- Corrupted and lost frames **must be resent**
- Sender maintains a copy of sent frame and starts a **timer**
- If timer expires and no ACK received, resend
- ACK frames can also be corrupted

Stop and Wait ARQ

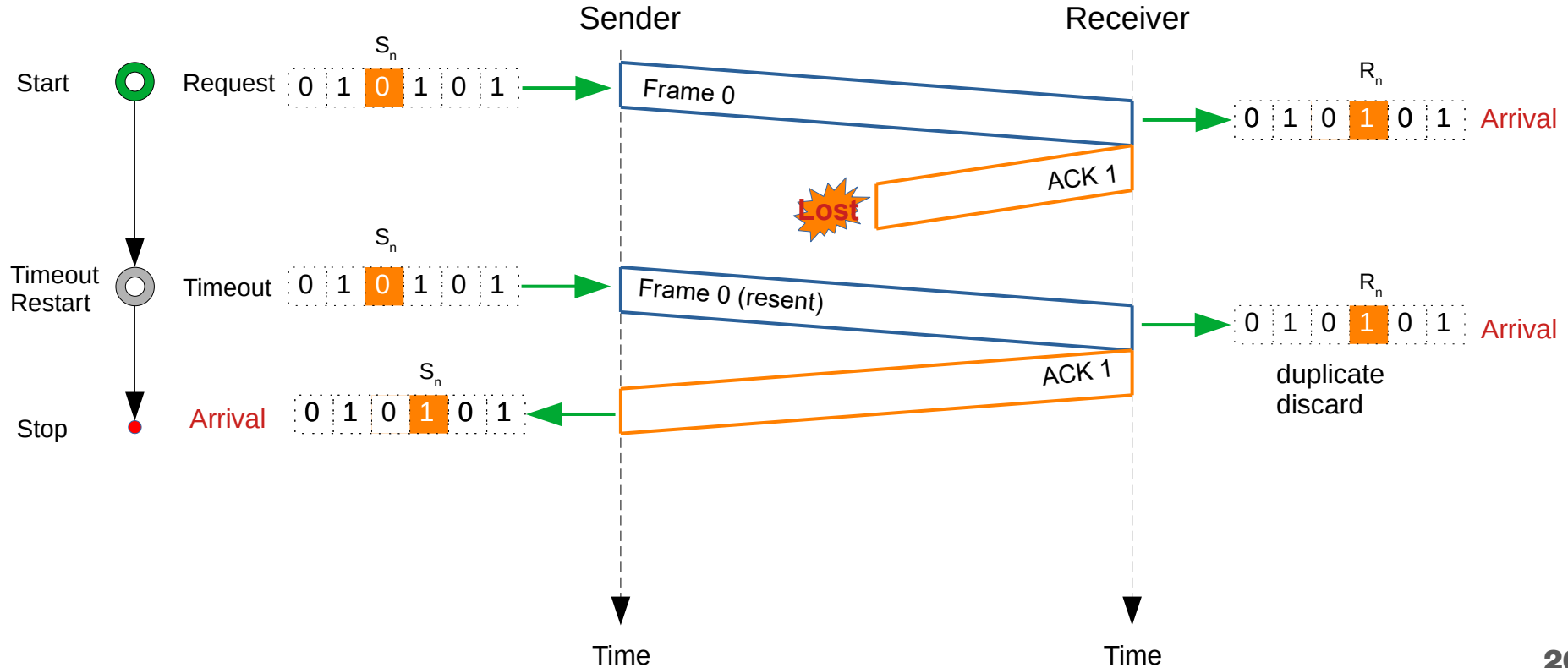
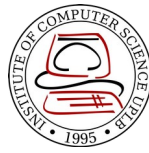


- Sequence numbers
 - Select the smallest range that provides unambiguous communication; also to minimize frame size
 - If m bits long, sequence number start from 0 to $2^m - 1$ and then are repeated
- Acknowledgment numbers
 - Announces the next sequence number expected
- In Stop-and-Wait ARQ, the sequence number and acknowledgment number are based on modulo-2 arithmetic

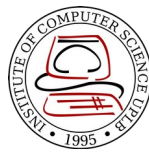
Stop and Wait ARQ



Stop and Wait ARQ

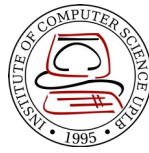


Stop and Wait ARQ



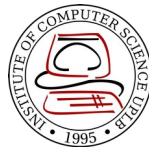
- Inefficient if channel is thick and long: large bandwidth and round-trip delay is long
- Recall: bandwidth-delay product (BDP) – volume of pipe in bits
- BDP is a measure of the number of bits we can send out of our system while waiting for news from the receiver

Go-Back-N ARQ



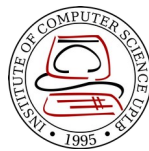
- Multiple frames must be in transit while waiting for ACK - **pipelining**
- We keep a copy of the frames in transit until acknowledgment arrives
- Sequence numbers are **modulo 2^m** , where **m** is the size of the sequence number field in bits
- **ACKs are cumulative** – more than one frame can be acknowledged by a single ACK

Go-Back-N ARQ

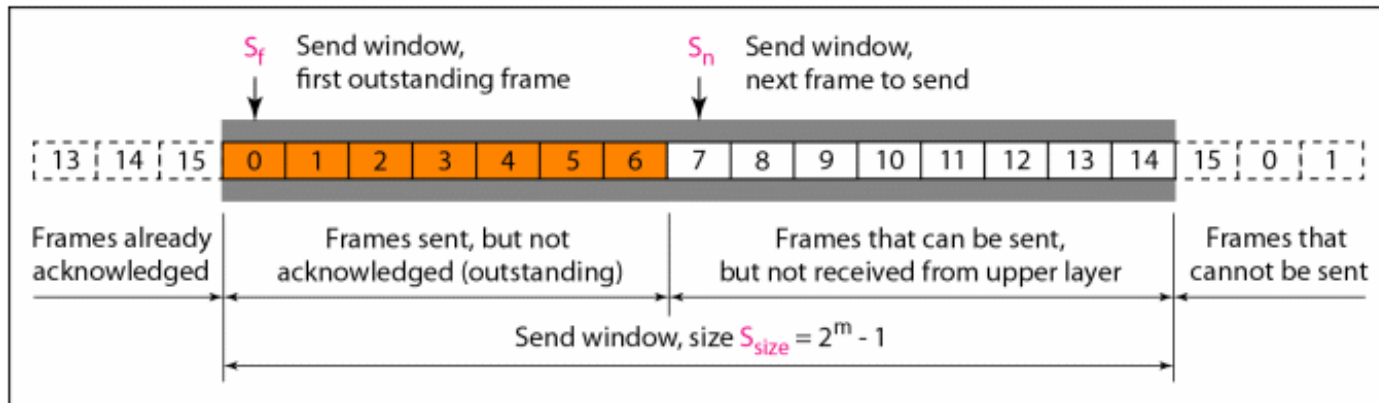


- Sliding Window
 - Abstract concept that defines the range of sequence numbers that is the concern of the sender and receiver
- Send Sliding Window
 - Imaginary box covering the sequence numbers of the data frames which can be in transit
 - Can slide one or more slots when a valid acknowledgment arrives
- Receive Sliding Window
 - Imaginary box of size 1
 - The window slides when a correct frame has arrived; sliding one slot at a time

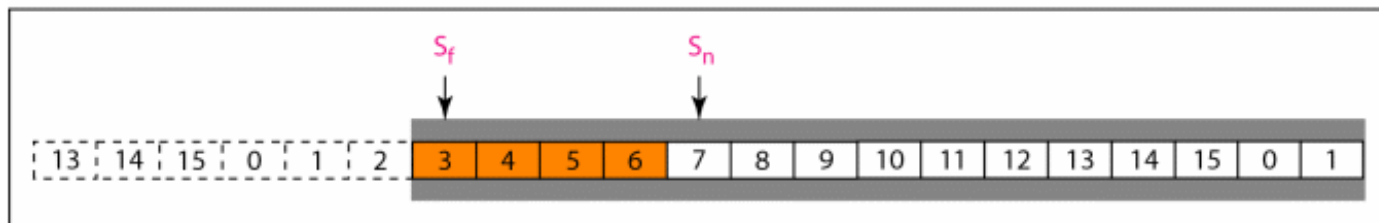
Go-Back-N ARQ



Send window for Go-Back-N ARQ; S_f , S_n , S_{size}



a. Send window before sliding

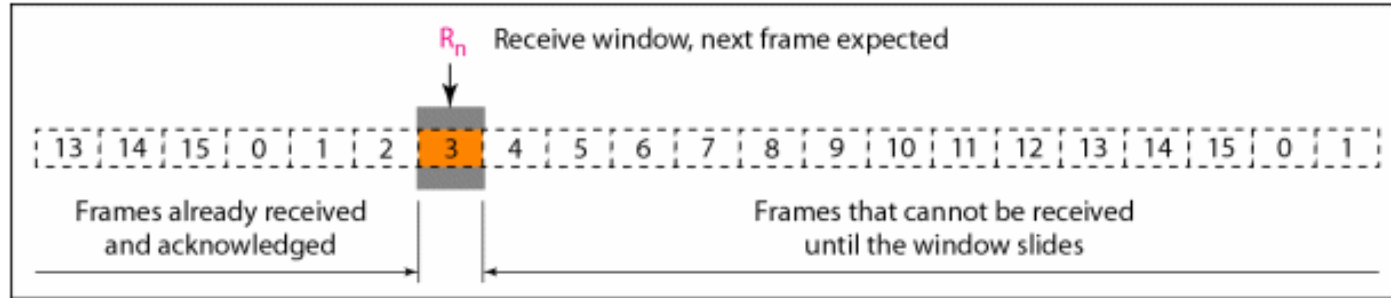


b. Send window after sliding

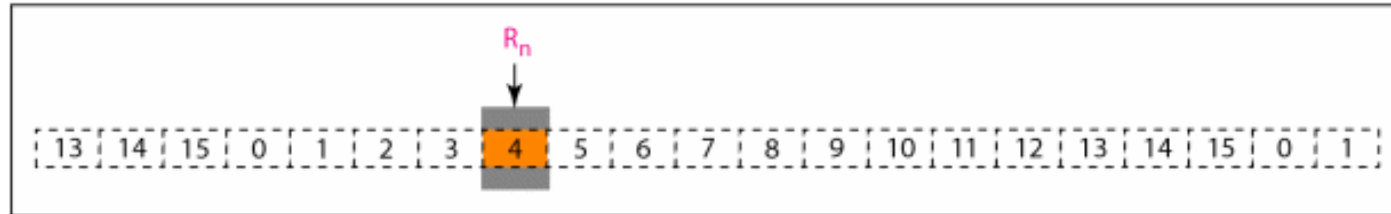
Go-Back-N ARQ



Receive window for Go-Back-N ARQ; R_n

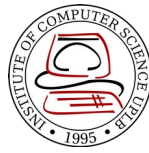


a. Receive window



b. Window after sliding

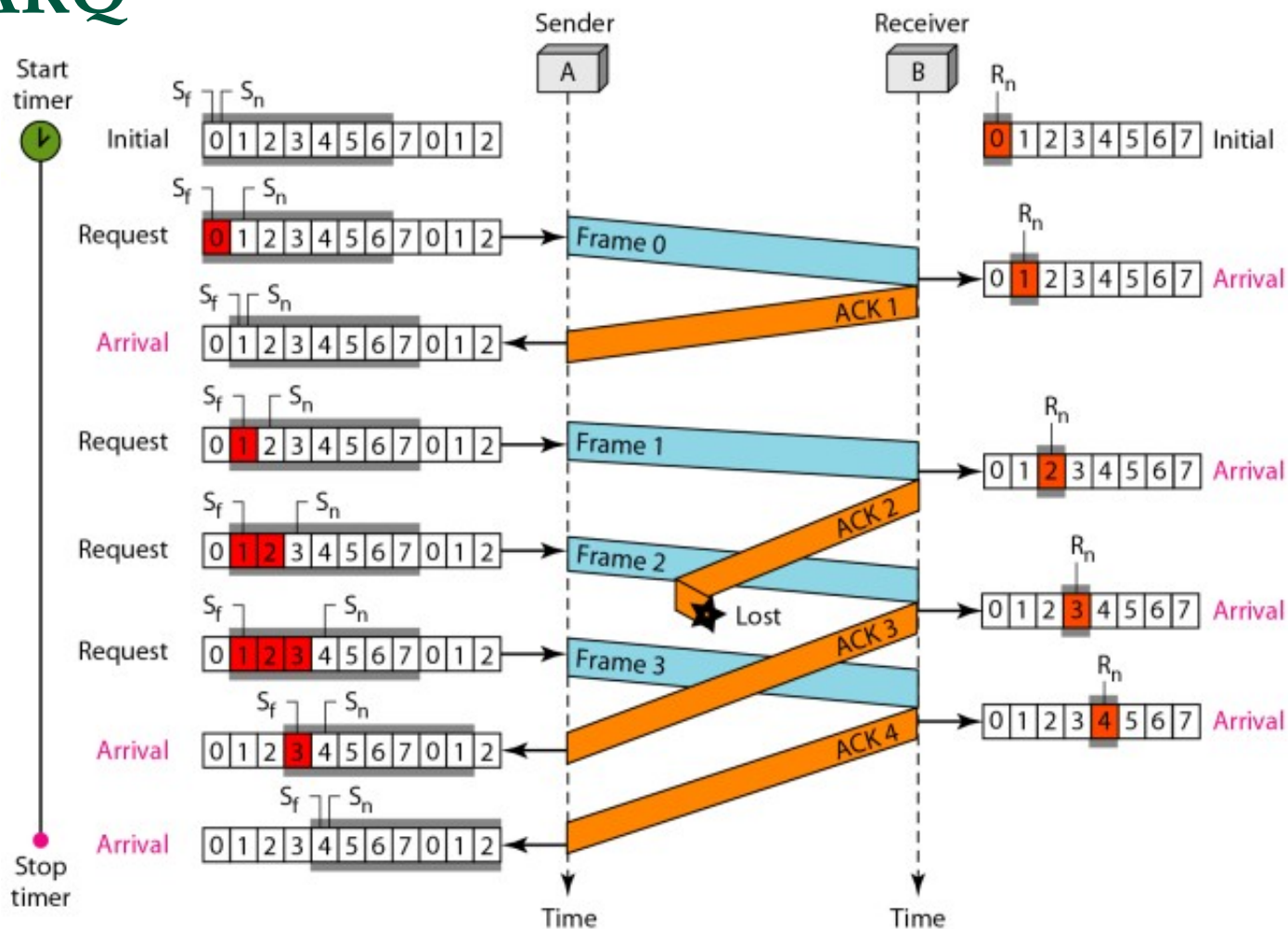
Go-Back-N ARQ



- Only **one timer** is used since the first outstanding frames always expires first
- **Silence of receiver** cause the timer of the unacknowledged frame at the sender site to expire
- Sender **re-sends all outstanding frames**
- Ex. Sender has already sent frame 6, but timer for frame 3 expires. Sender must resend 3,4,5,6

Go-Back-N ARQ

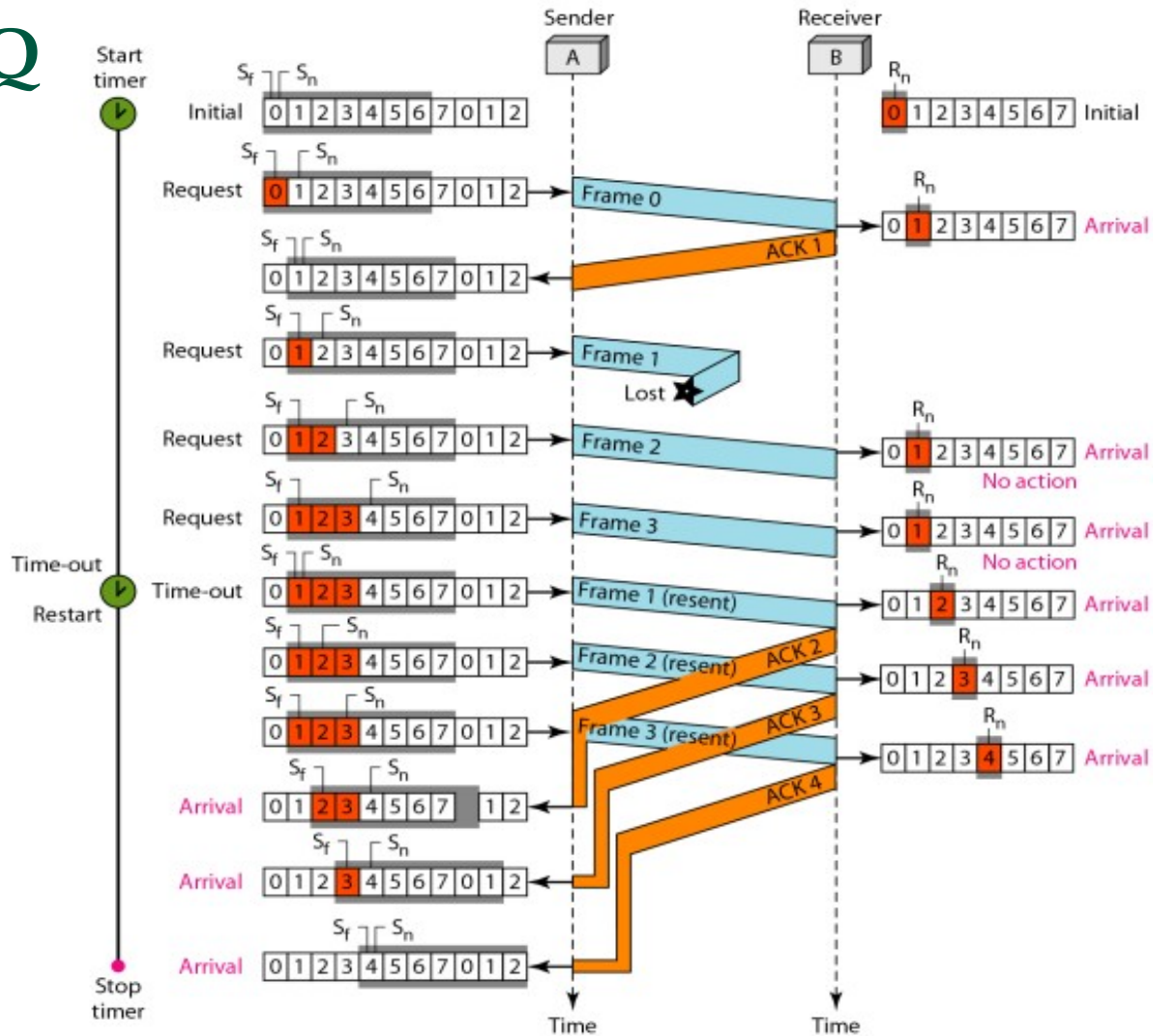
Forward channel is reliable



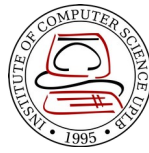
Go-Back-N ARQ

Lost Frames

Stop-and-Wait ARQ is a special case of GBN ARQ with send window size of 1

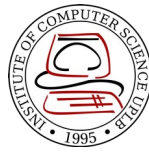


Selective Repeat ARQ

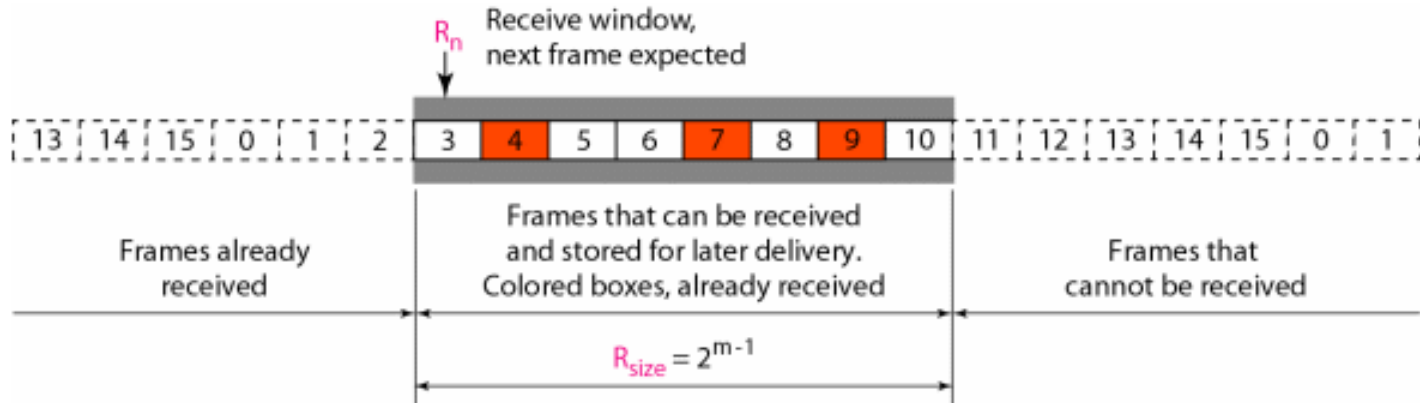
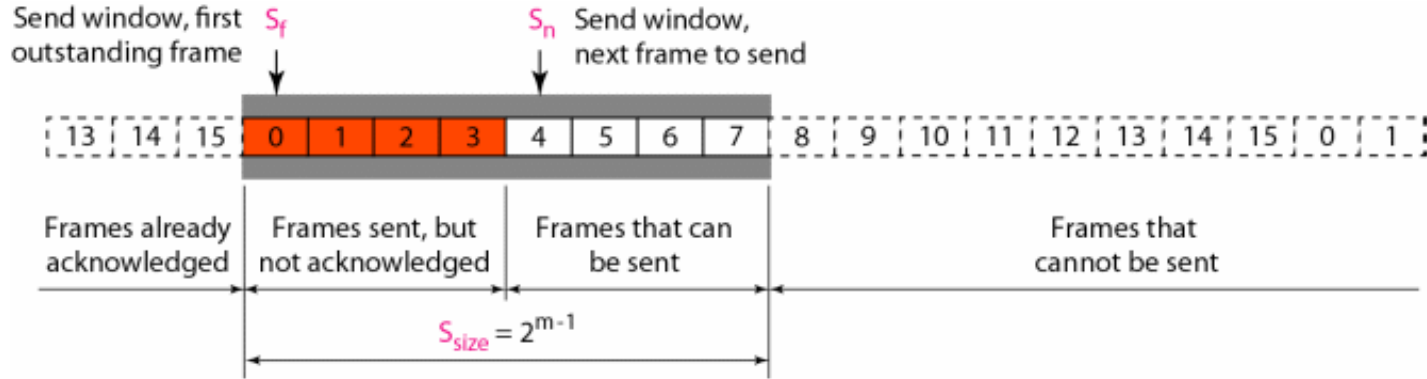


- Only damaged frames are resent
- More efficient for noisy links
- Processing at the receiver is more complex
- Send window max size is 2^{m-1} ; same size in receive window; smaller than GBN
 - Less efficient but fewer duplicate frames

Selective Repeat ARQ

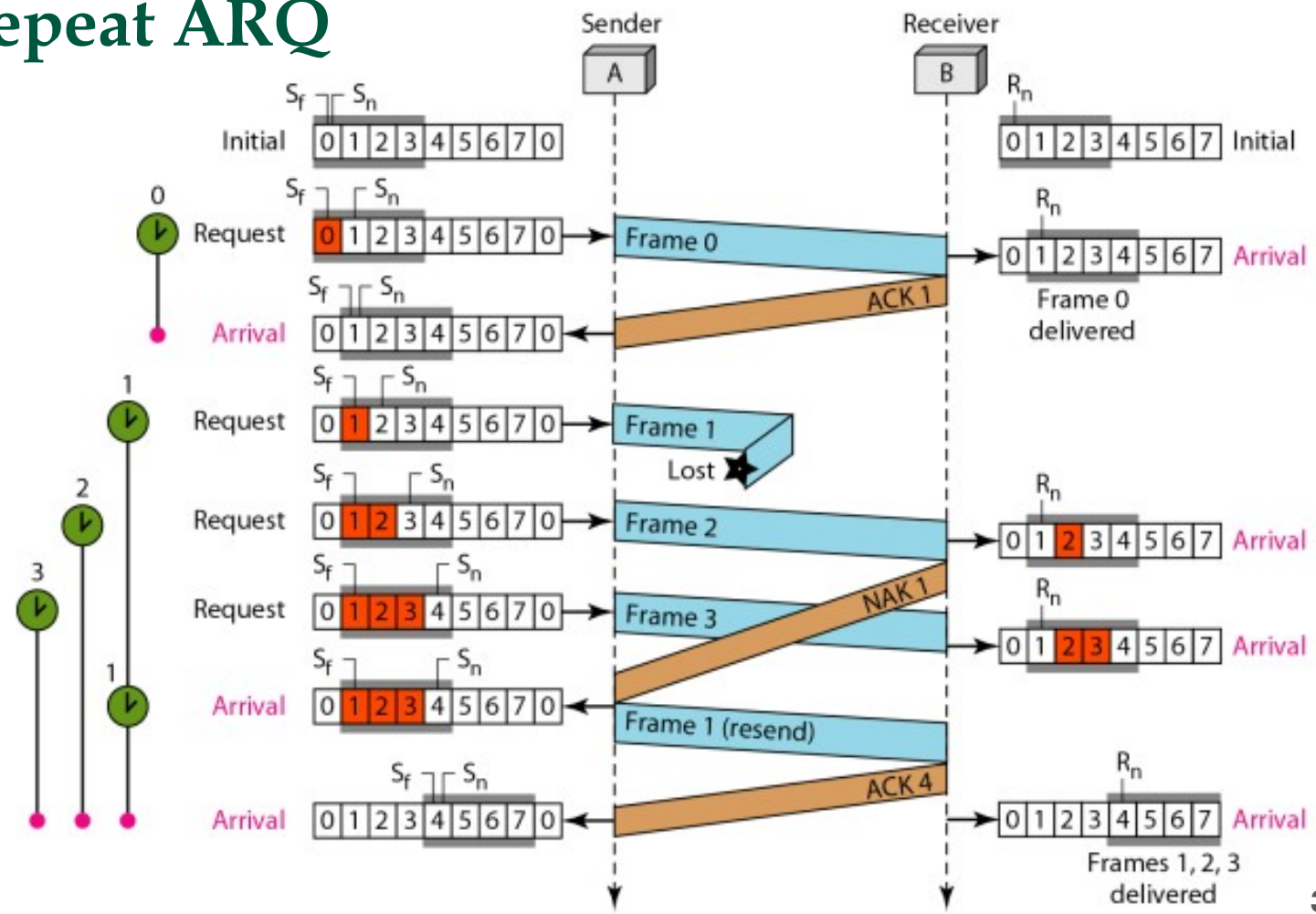


Send Window

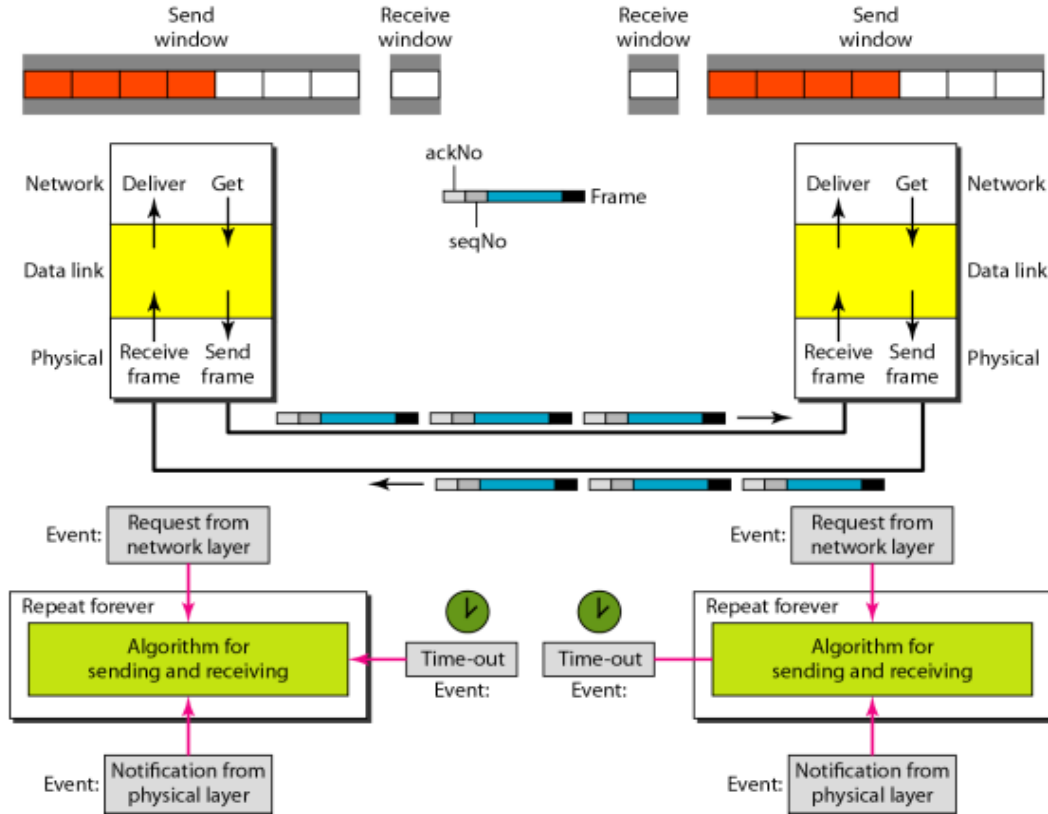


Receive Window

Selective Repeat ARQ



Piggybacking in Go-Back-N ARQ



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.